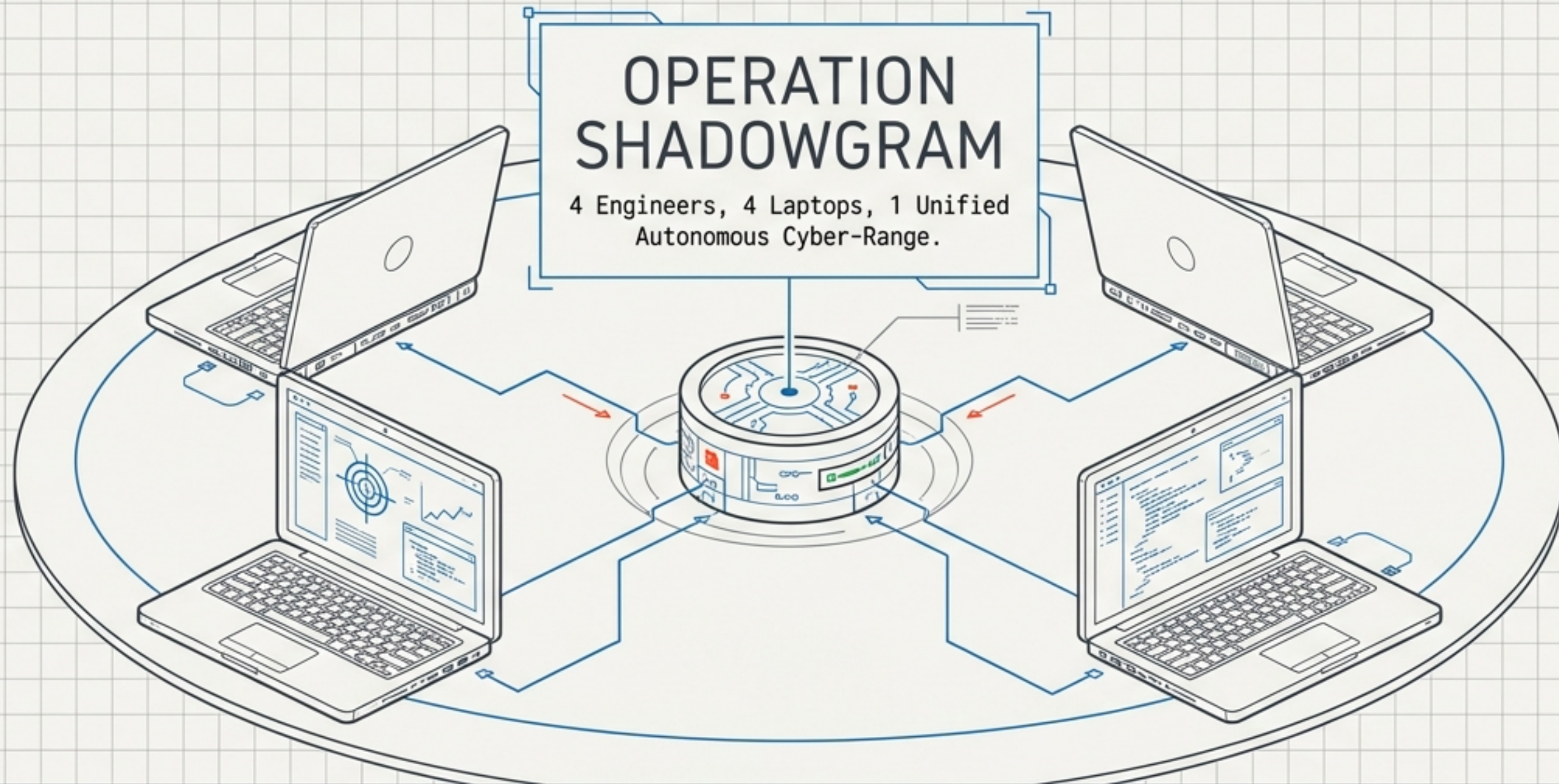
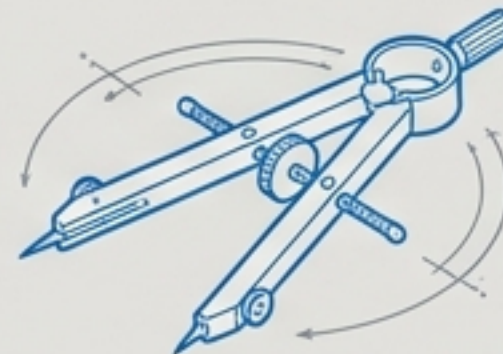




CORE OBJECTIVE

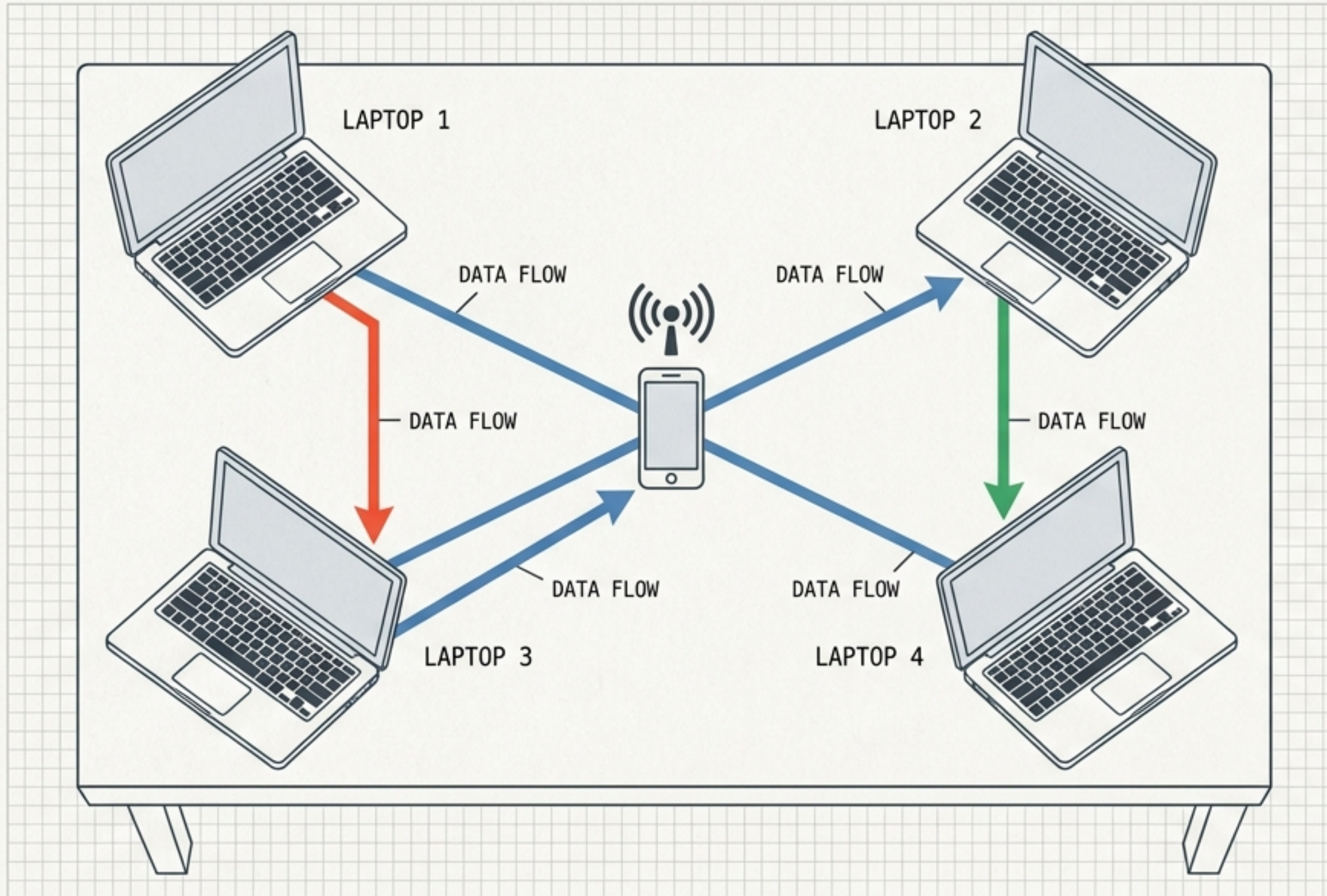
Execute an airtight, live-fire cybersecurity demonstration with zero pre-recorded elements or mocked APIs.



THE PARADIGM

A fully functional, localized battle-station setup designed to process adversarial bot swarms and isolate threats in real-time, directly in front of the evaluation panel.

PHYSICAL ARCHITECTURE OF THE LIVE CYBER-RANGE



LAPTOP 1: THE ATTACKER

Red Team Swarm Runner

Launches 20 automated Playwright bot instances simultaneously.

LAPTOP 2: COMMAND COCKPIT

ShadowGram Server & Visualization

Runs FastAPI, SQLite, and 60 FPS WebGL 3D dashboard.

LAPTOP 3: TARGET PORTAL

AthenaPay Loan Application

Clean micro-lending web interface for live judge manual entry.

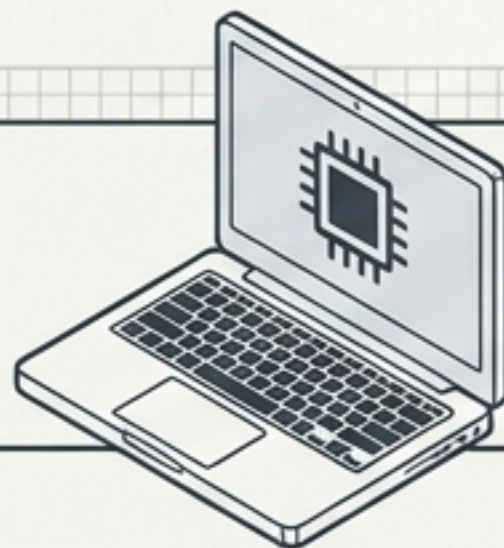
LAPTOP 4: COMPLIANCE

Officer's Legal Terminal

Houses the Why Card, 1-click quarantine trigger, and SAR exporter.

STRATEGIC HARDWARE AND COMPUTE ALLOCATION

HIGH-COMPUTE TIER (GAMING)



OPERATIONAL WORKLOAD

STATION 1 (ATTACKER)

Drives 20 parallel Playwright browser contexts via 1 optimized Chromium instance.

STATION 2 (COCKPIT)

Hosts FastAPI backend, ONNX 2D-CNN, Sentence-Transformers, and renders 3D WebGL at 60 FPS.

MEMORY FOOTPRINT

ST1_RAM: ~600MB (Optimized Contexts)

ST2_RAM: GPU Accelerated Heavy Workload

STANDARD TIER (NORMAL)



STATION 3 (PORTAL)

Serves standard Next.js 3-step loan web application for judge input.

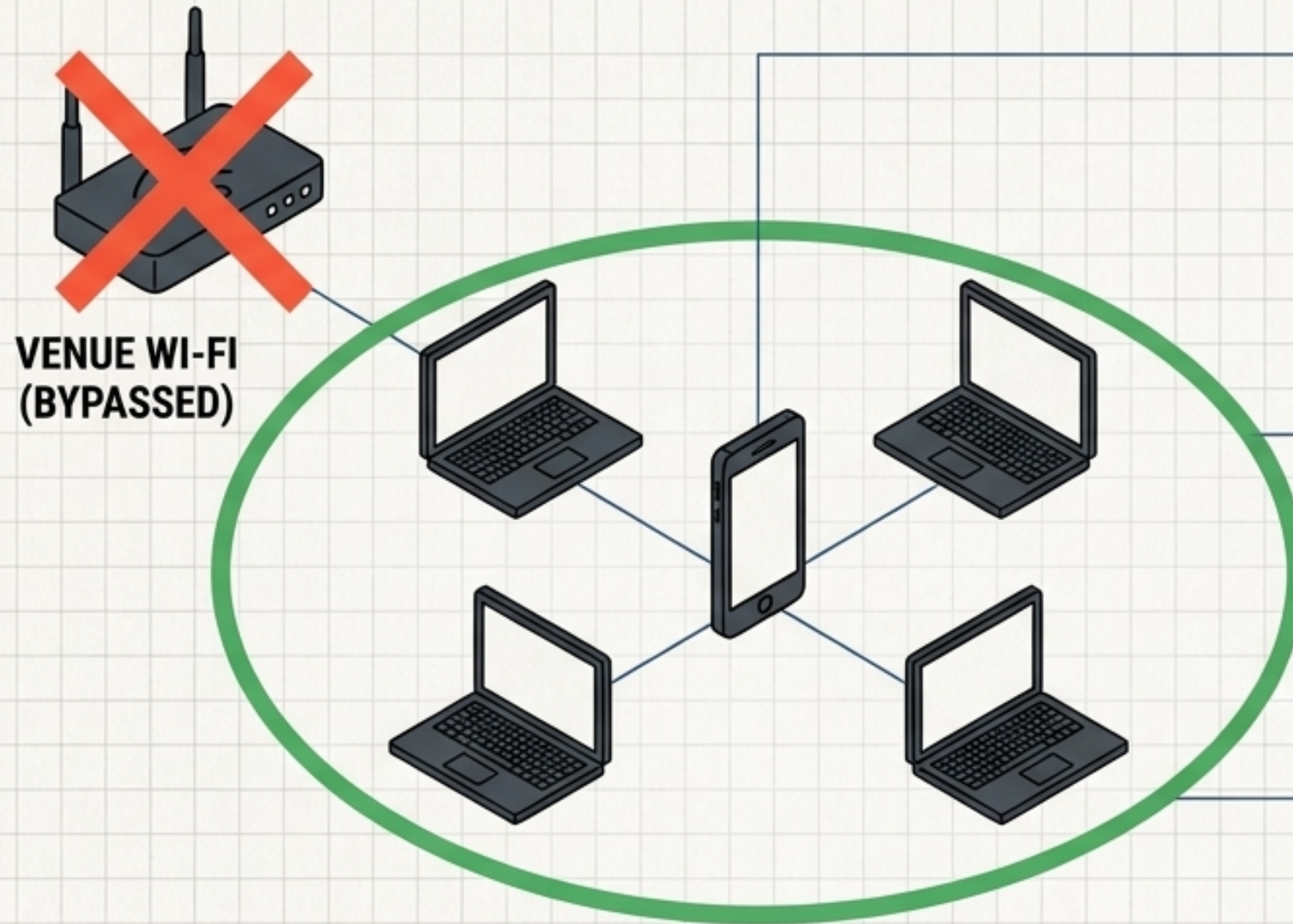
STATION 4 (COMPLIANCE)

Renders compliance data tables and processes PDF generation via ReportLab.

ST3_RAM: <200MB

ST4_RAM: <300MB (LLM offloaded to NVIDIA Cloud)

ZERO-CLOUD RESILIENCE AND LOCAL LAN TOPOLOGY



RULE 1: FASTAPI HOST BINDING

Localhost is strictly forbidden to ensure peer connectivity across the bubble.

```
uvicorn --host 0.0.0.0 --port 8000
```

RULE 2: DYNAMIC IP TARGETING

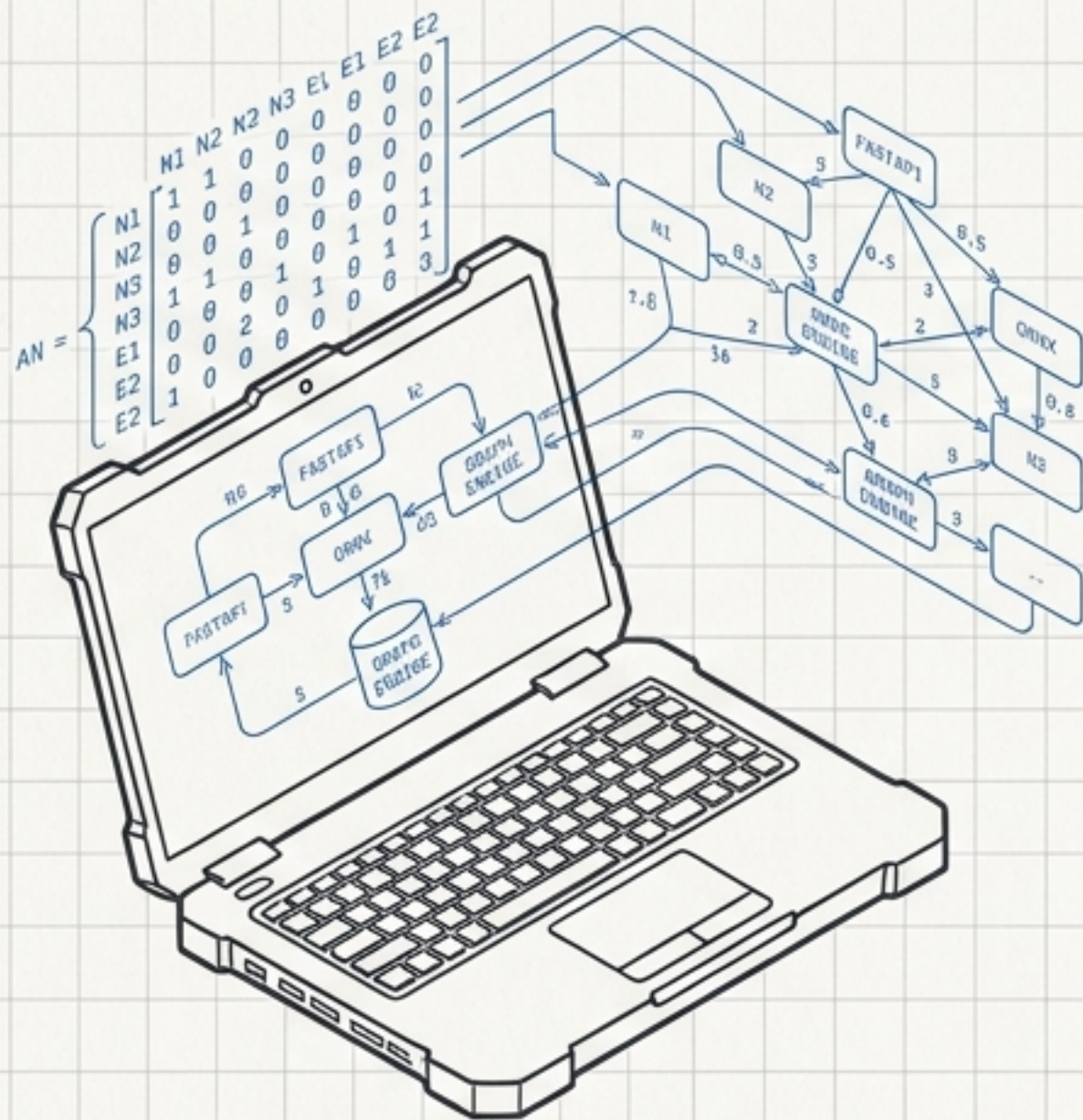
Server explicitly identifies local IP address. All telemetry hooks target this absolute address.

```
hostname -I -> 192.168.43.2
```

RULE 3: CROSS-PLATFORM NATIVE EXECUTION

Total prohibition of Docker to bypass Windows WSL2 bridge NAT packet drops. Pure Python venv and native Node.js deployed.

OPERATOR DOSSIER: PRINCIPAL SYSTEMS ARCHITECT



OPERATOR: PARADOXPETE
STATION: LAPTOP 2 (COCKPIT)
HARDWARE: GAMING RIG

CORE ENGINEERING DELIVERABLES	
Backend Architecture	Architects the core FastAPI backend and asynchronous WebSocket dispatcher (backend/main.py).
Graph Mathematics	Implements ShadowGraphEngine utilizing Louvain modularity clustering and dynamic edge weighting.
Kinetic Classification	Integrates pre-exported 2D-CNN ONNX model for cursor jerk classification.
Semantic Vectors	Deploys sentence-transformers/all-MiniLM-L6-v2 for dense semantic vector embeddings.

PERFORMANCE METRIC: SUB-10 MILLISECOND
CLASSIFICATION RESPONSE TIME

Lat: <10ms | S_comp >= 0.78

OPERATOR DOSSIER: VISUAL COMMAND CENTER LEAD



OPERATOR: AISWARYA KALLAYIL RAJESH
STATION: LAPTOP 2 (MAIN DISPLAY)
HARDWARE: GAMING RIG

CORE ENGINEERING DELIVERABLES

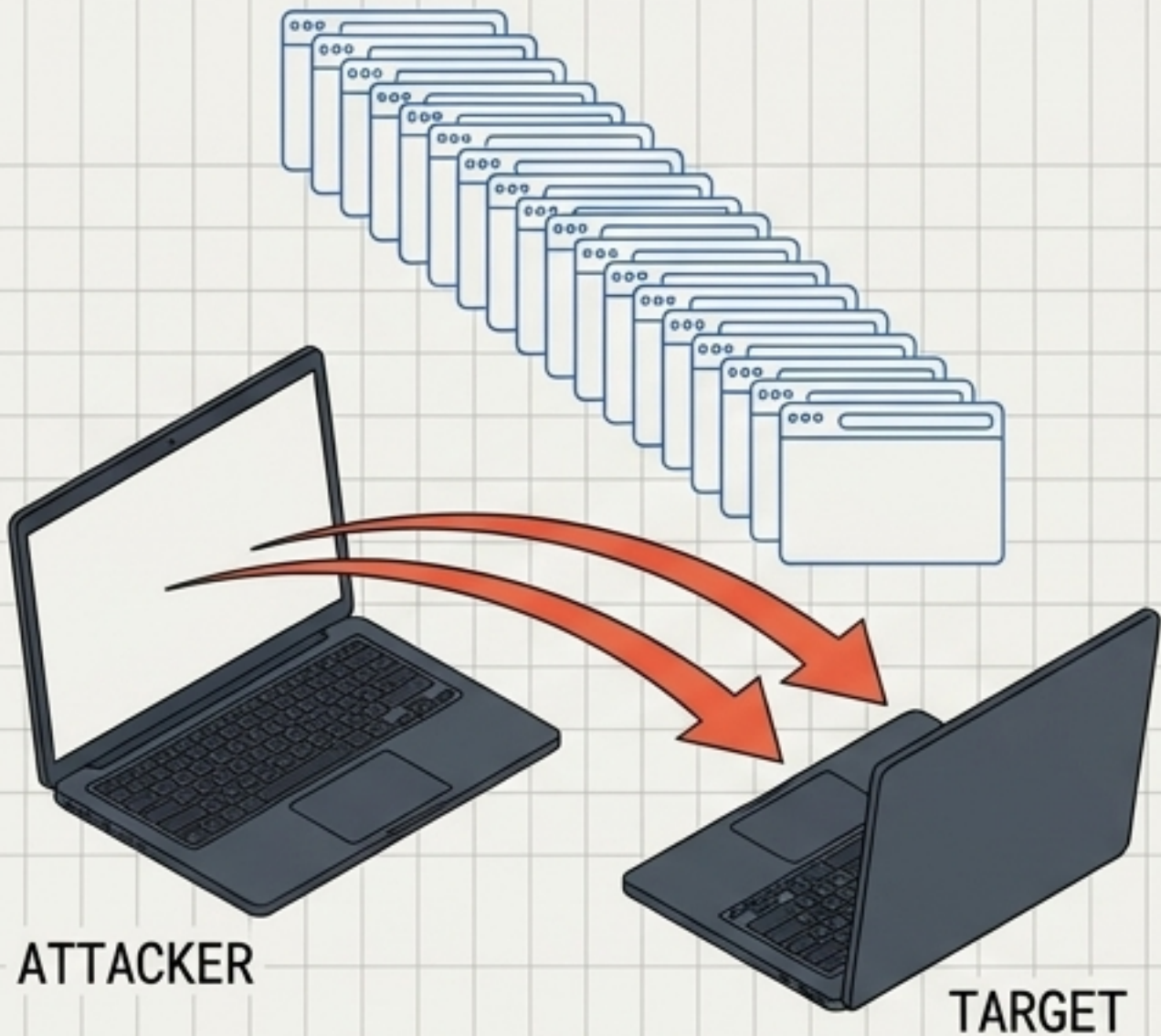
Deliverable	Description
Frontend Architecture	Develops Next.js 14 frontend using Tailwind CSS and custom editorial data-visualization styling.
WebGL Integration	Engineers react-force-graph-3d (Three.js WebGL) integration with camera rotation and 2D fallback toggles.
Data Transparency UI	Builds the interactive Why Card modal to render real-time factual telemetry (arrival delta-t, LCS route overlap).
Action States	Wires the 1-Click Blast-Radius Quarantine and Deploy Swarm Simulation command triggers.

PERFORMANCE METRIC: HOLLYWOOD-GRADE 60 FPS
WEBGL RENDERING OF COMPLEX MATH

OPERATOR DOSSIER: ADVERSARIAL SWARM RUNNER

CORE ENGINEERING DELIVERABLES

Swarm Orchestration	Authors swarm_runner.py via Python Playwright async API to spawn 20 simultaneous bot sessions.
Synthetic Personas	Queries free NVIDIA NIM API (Llama-3.3-70B) for realistic, non-repeating applicant profiles.
Kinetic Simulation	Injects parametric cubic Bezier curves to simulate automated and repetitive cursor pathing.
Telemetry Instrumentation	Develops telemetry.js to capture keystroke flight/dwell intervals and throttled pointer coordinates.

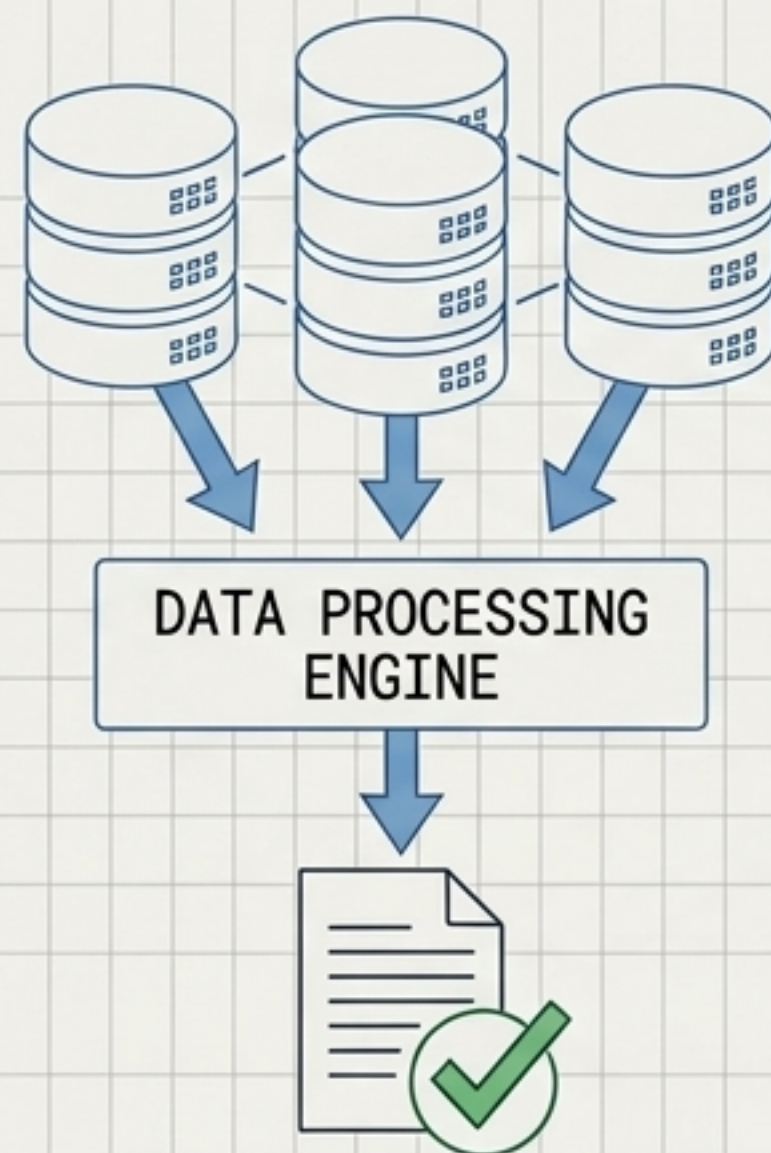


OPERATOR: ALAN E ALEXANDER
STATION: LAPTOP 1 & LAPTOP 3
HARDWARE: GAMING + NORMAL RIG

PERFORMANCE METRIC: STRICT ZERO-PII
ENFORCEMENT PROTOCOL

Streaming strictly numerical time deltas
and Cartesian coordinates to cockpit.

OPERATOR DOSSIER: DATA ENGINEERING & LEGAL COMPLIANCE



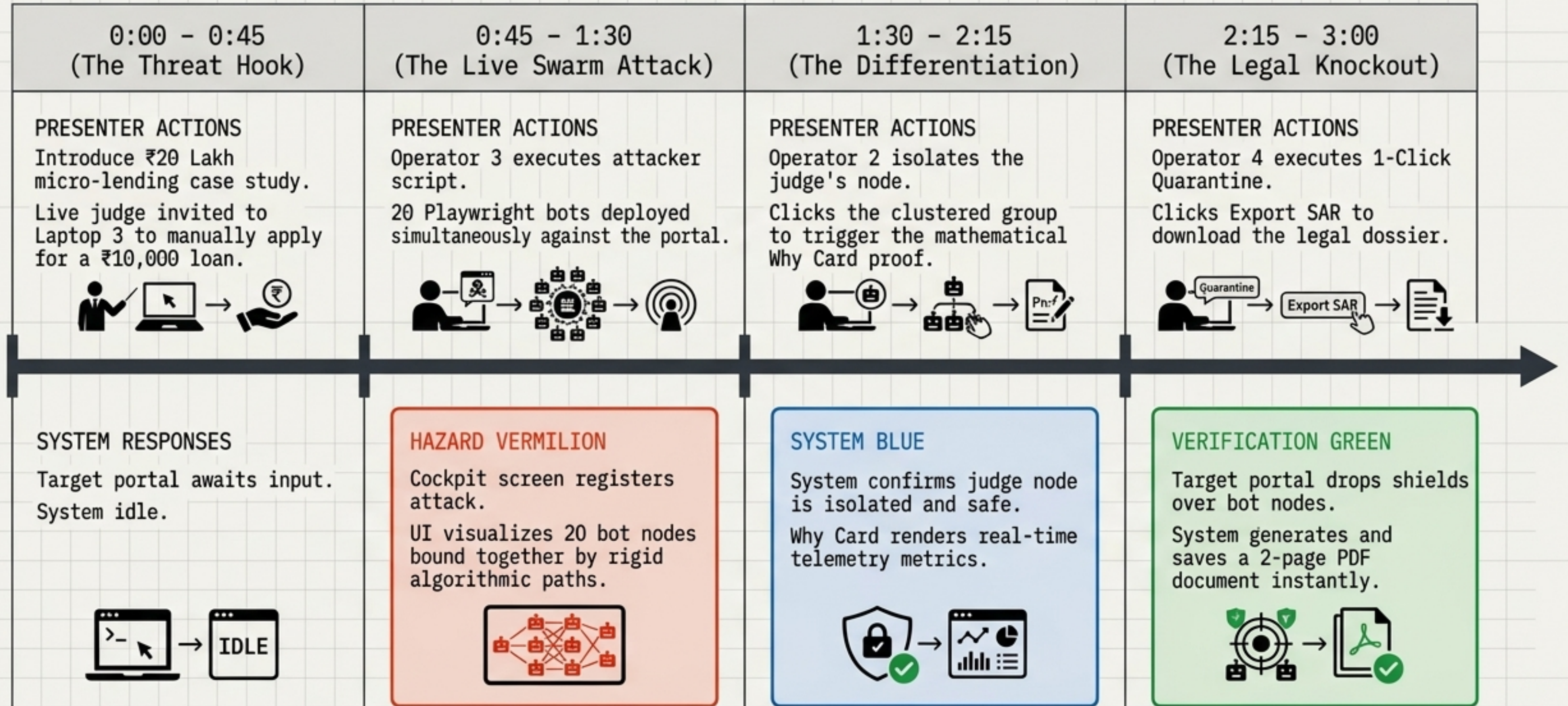
OPERATORS: NIHAD / ASHLIN
STATION: LAPTOP 4 (OFFICER)
HARDWARE: NORMAL RIG

CORE ENGINEERING DELIVERABLES

Deliverable	Description
Database Architecture	Defines SQLite schema (shadowgram.db) using SQLAlchemy for sessions, events, and clusters.
Legal Translation	Authors sar_generator.py calling NVIDIA NIM API to translate raw cluster metrics into CFPB compliant adverse action reasons.
Document Generation	Implements generate_sar_pdf.py using Python ReportLab to output a clean 2-page legal dossier.
Fraud Detection	Deploys kyc_scanner.py using PIL/OpenCV for lightweight synthetic ID detection.

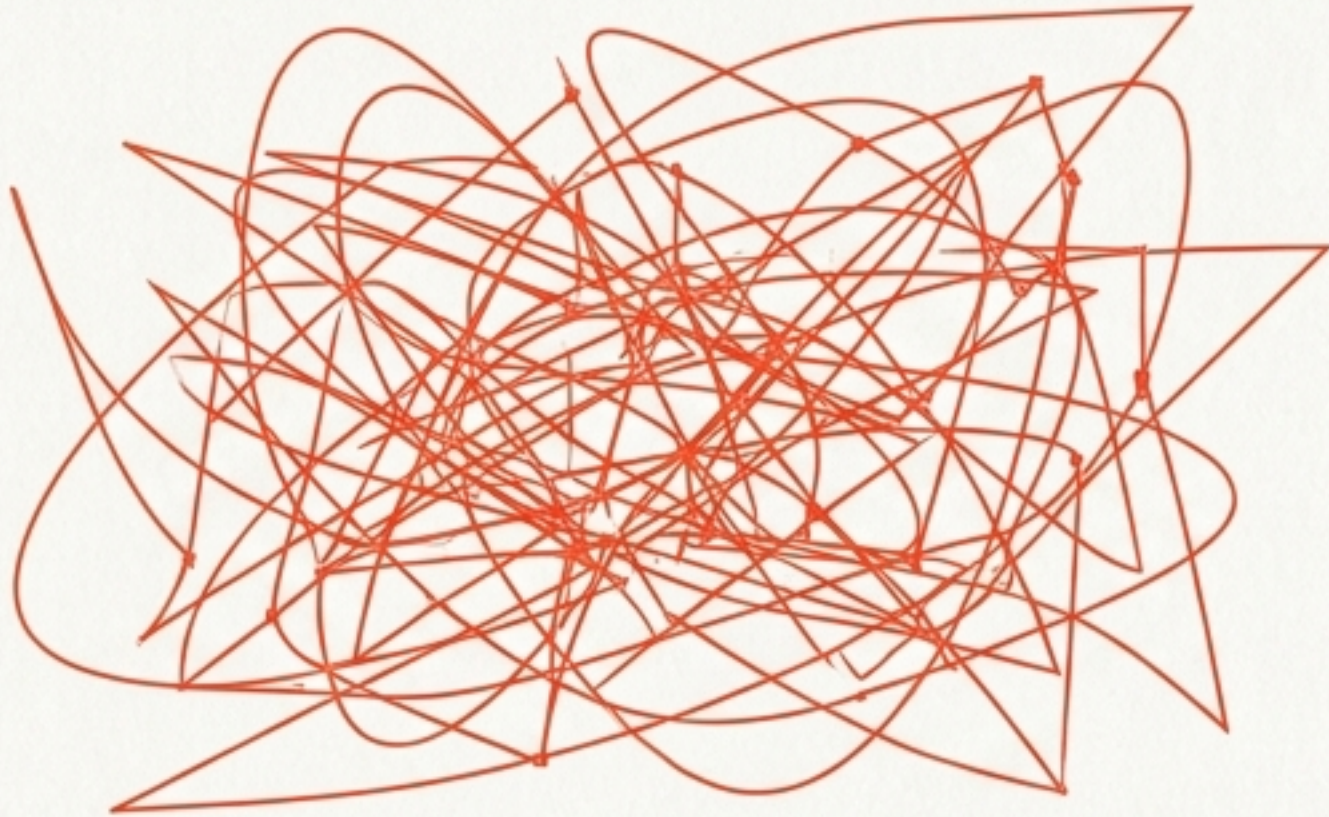
PERFORMANCE METRIC: SEAMLESS CONVERSION OF ADVERSARIAL
TELEMETRY INTO REGULATOR-READY SAR PDF

THE 180-SECOND LIVE PITCH CHOREOGRAPHY



THE GOLDEN TOUCH: LIVE JUDGE INTERACTION

THE BOT SWARM (AUTOMATED)



HIGH MODULARITY / CLUSTERED

Rigid geometric pathing detected. Synchronous flight/dwell intervals flagged.

THE LIVE JUDGE (ORGANIC)



LOW MODULARITY / ISOLATED ($Q < 0.15$)

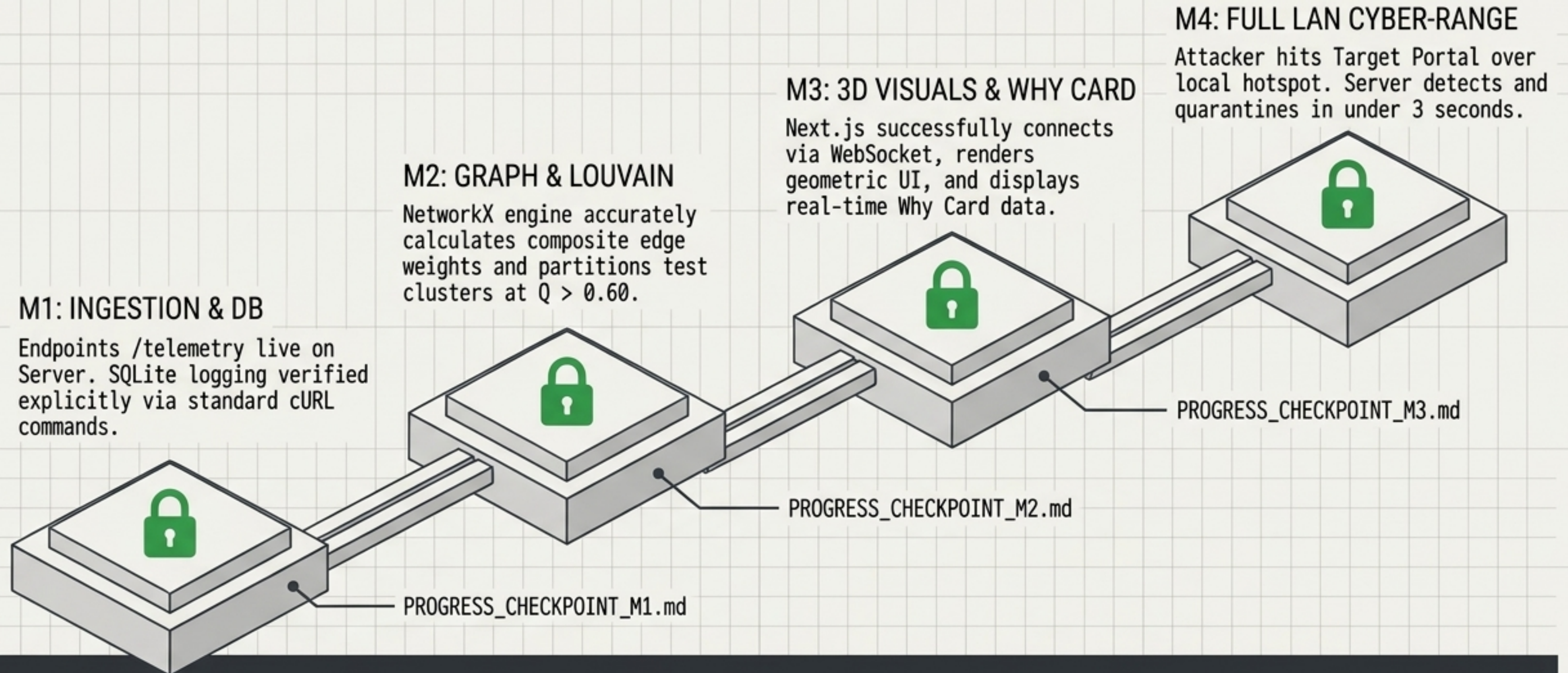
Organic typing cadence and natural mouse tremors mathematically distinct.

OVERCOMING SKEPTICISM: Handing the keyboard to a judge destroys the illusion of pre-baked slides.

REAL-TIME MATH: Louvain modularity instantly flags the rigid perfection of the 20 bots while maintaining the human safe zone.

THE UX REALITY: The evaluation panel experiences the defense not as a block diagram, but as an active response to their physical touch.

ANTI-HALLUCINATION ENGINEERING & MILESTONE QUALITY-GATES



THE CHECKPOINT RULE: Absolute zero API schema drift. Every member builds against frozen quality gates, verifying locally before clean merges.